

August 5, 2026

Editor-in-Chief
Empirical Software Engineering

Dear Editor,

Subject: Submission of manuscript “Trust Assumptions in Self-Healing LLM Agent Frameworks: An Empirical Code Analysis of Recovery-Path Provenance Gaps”

We are pleased to submit the above-referenced manuscript for consideration by *Empirical Software Engineering*.

Summary. This paper presents an empirical study of trust-boundary design flaws in self-healing LLM agent frameworks. We employ a three-phase methodology addressing three research questions: (RQ1) How prevalent are recovery-path provenance gaps across production agent frameworks?—a multiple-case study of six frameworks (Reflexion, MetaGPT, Aider, OpenHands, AutoGen, LangGraph) with grounded-theory coding; (RQ2) How exploitable and severe are these flaws?—controlled experiments on a white-box testbed (100 trials per configuration, real LLM APIs), cross-framework validation on LangGraph, and a confirmed end-to-end exploit on Reflexion’s actual retry path (0.52 ASR, $p < 0.001$); (RQ3) How effective is the defense proposed in our companion paper, and under what conditions does it degrade?—naive-baseline comparison, cross-model generalization (open-weights and closed-weight models), defense-aware attacker analysis, and defense-cost measurement.

Contribution type. This is an empirical software engineering contribution. The paper uses established empirical methodologies: multiple-case study design (Runeson & Höst 2009), grounded-theory coding (Glaser & Strauss 1967), controlled experiments with statistical hypothesis testing (Wilson confidence intervals, Fisher exact tests with Bonferroni correction), inter-annotator agreement (Cohen’s κ), and threats-to-validity analysis. The paper reports both confirmatory and non-confirmatory results, including under-powered experiments reported honestly as suggestive rather than confirmatory.

Disclosure of companion manuscript. We wish to transparently disclose that the authors have prepared a companion manuscript submitted to the *Journal of Systems and Software* (JSS), titled “Recovery-Path Trust Boundaries in Self-Healing LLM Agent Systems: Architectural Analysis, Empirical Evidence, and Design Patterns.” The two manuscripts are complementary and non-overlapping in their primary contributions:

- **This manuscript (EMSE)** contributes the *empirical evaluation*: the multiple-case study, the controlled experiments with statistical tests, the confirmed end-to-end exploit, the cross-model generalization, and the defense-aware attacker analysis.
- **The companion manuscript (JSS)** contributes the *architectural design*: the formalization of the trust-escalation trampoline anti-pattern, the two design patterns (Provenance Tagging + Privilege Downgrade Invariant), and a concise validation of implementability.

The two manuscripts cite each other and are each self-contained: a reader of either paper can understand the full contribution without accessing the other. We have taken care to ensure that textual overlap between the two manuscripts is minimal (estimated $<15\%$); each manuscript uses

its own language to describe the shared system and attack surface.

We are happy to provide the companion manuscript as a confidential attachment for the editor’s review, should that be helpful for verifying non-overlap. We respectfully request that the companion manuscript not be shared with reviewers, to preserve the double-blind integrity of both submissions.

Fit with EMSE. The manuscript fits EMSE’s scope in empirical software engineering: it uses established empirical methodologies (multiple-case study, grounded theory, controlled experiments with statistical tests), studies a population of real-world software artifacts (production agent frameworks with combined $\sim 105k$ GitHub stars), and reports findings with appropriate statistical rigor including honest reporting of under-powered results. The architectural design patterns evaluated in RQ3 are proposed in the companion JSS paper; this paper’s contribution is their empirical evaluation.

We confirm that this manuscript has not been published and is not under consideration elsewhere. We look forward to the editors’ and reviewers’ feedback.

Sincerely,

Anonymous Author(s)
(for double-blind review)